

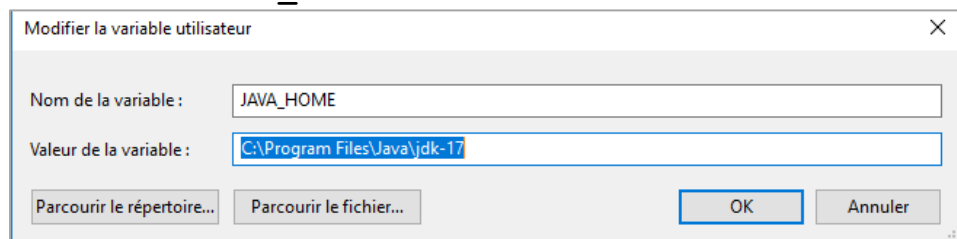


TP05 : PySpark

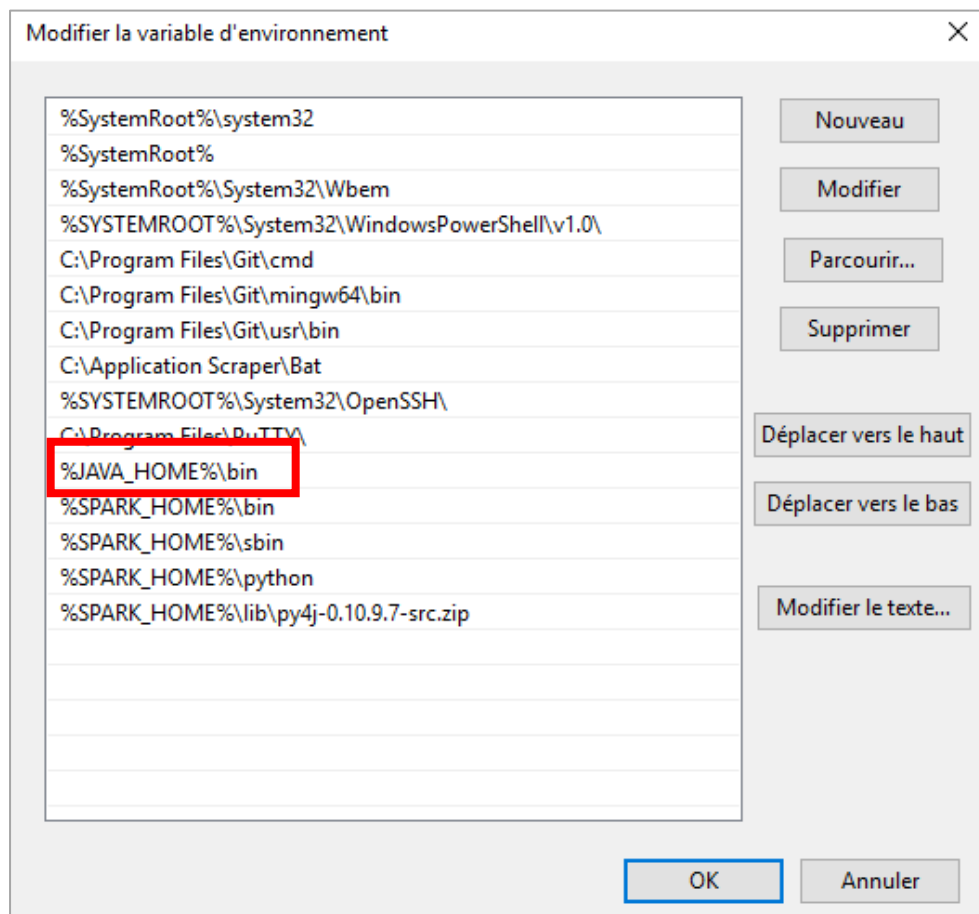
Prérequis :

1. Installer JDK : version 8, 11 ou 17 (<https://www.oracle.com/java/technologies/downloads/>)
2. Modifier les variables d'environnement

- Créer la variable **JAVA_HOME**



- Modifier la variable **PATH**

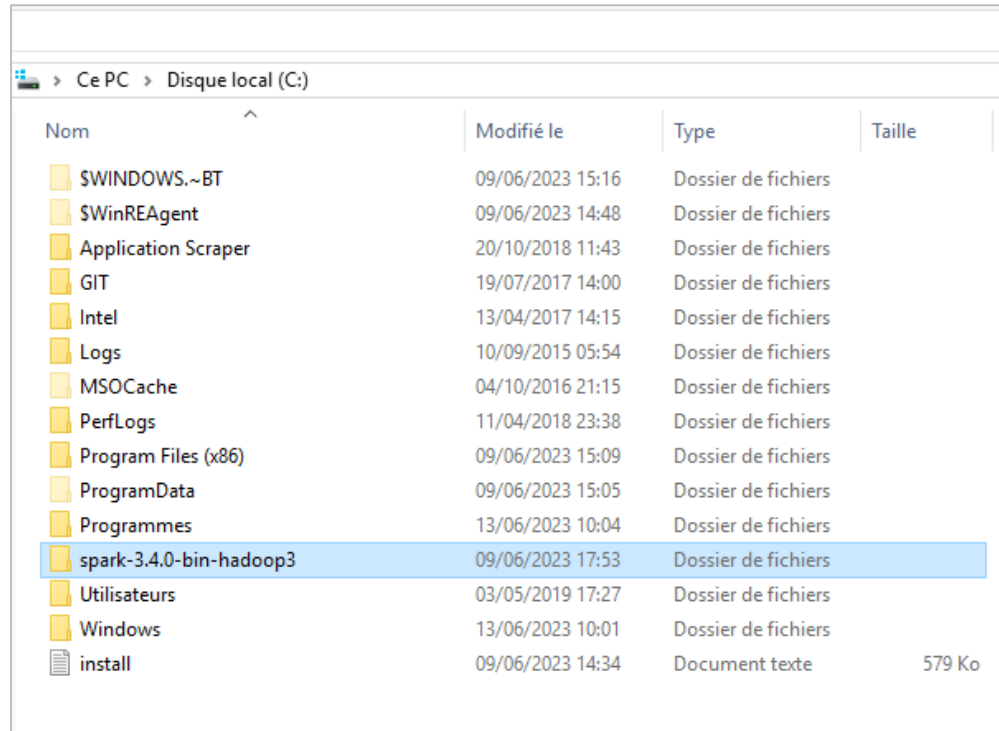


3. Installer Spark
 - Télécharger la version 3.4.0 (<https://spark.apache.org/downloads.html>)



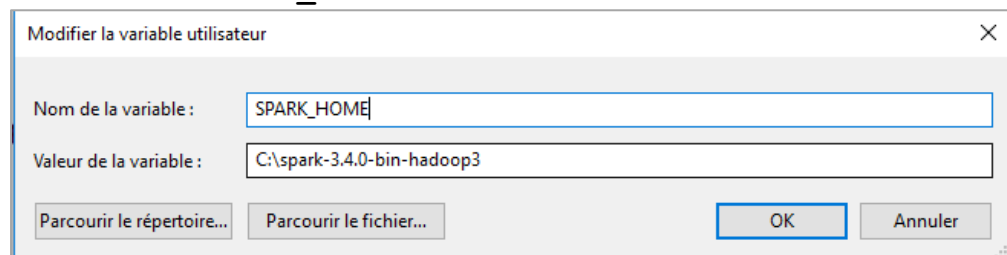
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- Extraire l'archive dans **C :** /

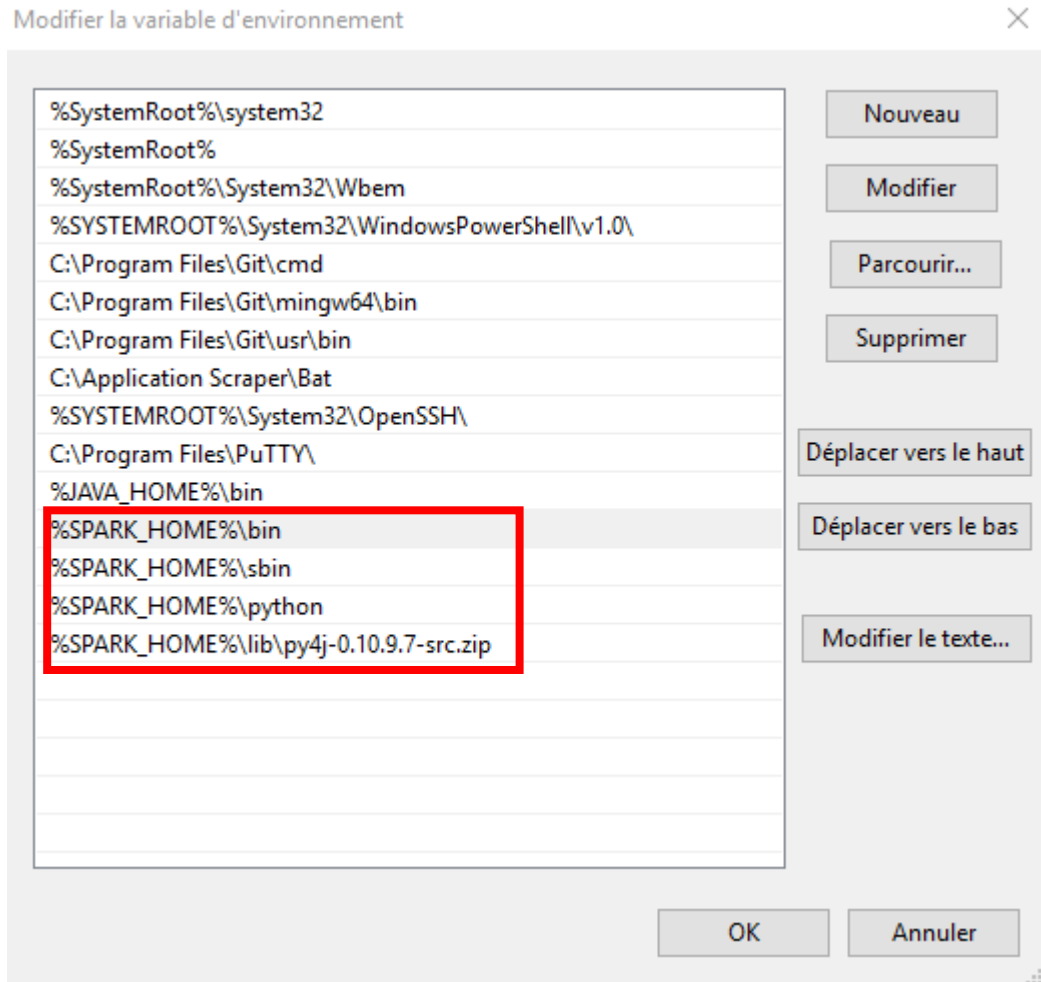


4. Modifier les variables d'environnement

- Créer la variable **SPARK_HOME**



- Modifier la variable **PATH**

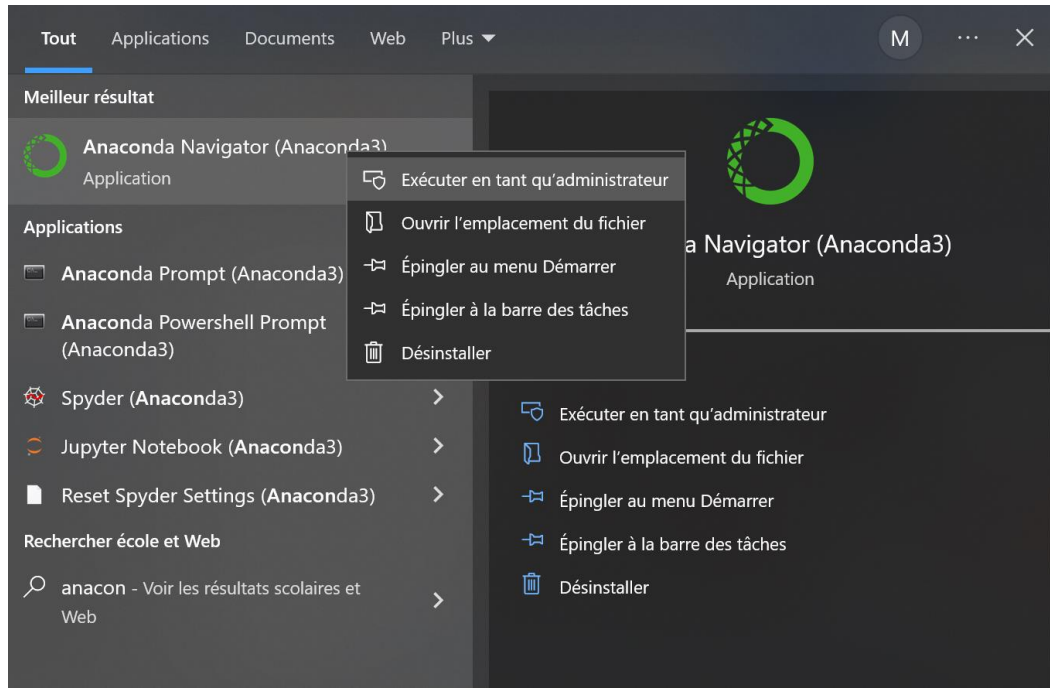


5. Installer la dernière version d'anaconda (<https://www.anaconda.com/download>)

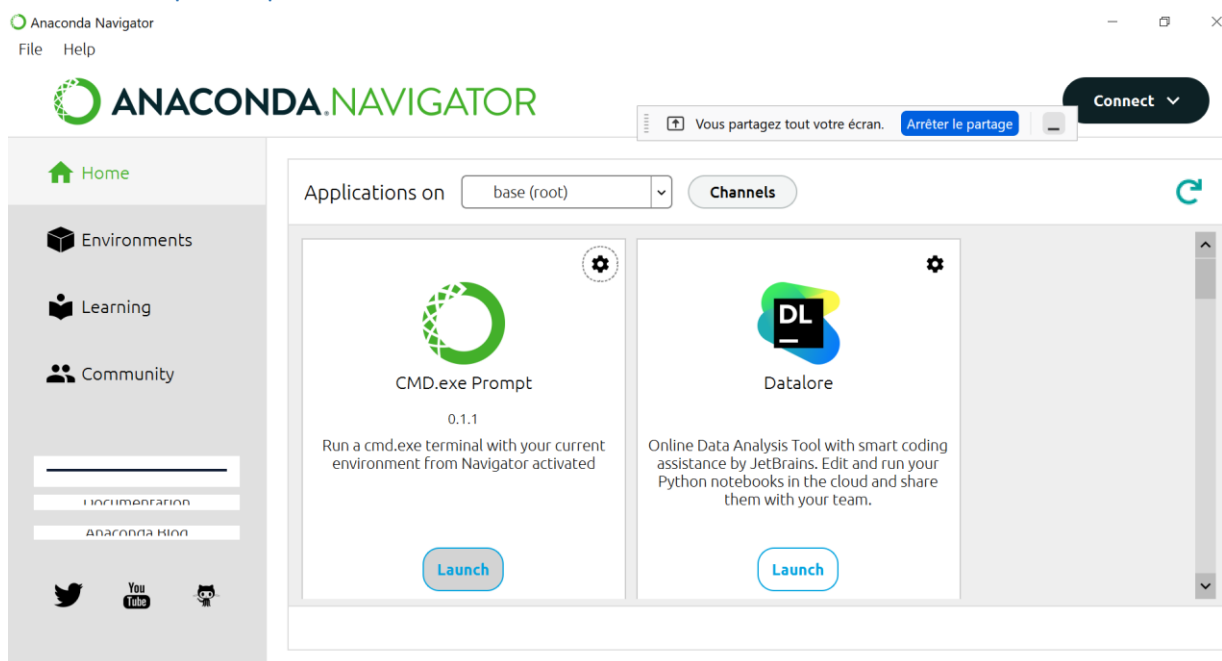


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Lancer Anaconda en tant qu'Administrateur



Lancer le prompt



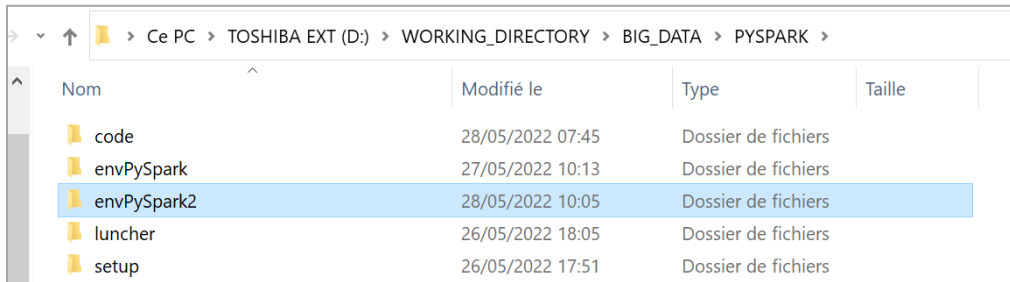


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```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [version 10.0.19043.1706]
(c) Microsoft Corporation. Tous droits réservés.

(base) C:\Users\mariendiaye>
```

Créer un dossier nommé « envPySpark2 » dans le répertoire de votre choix



Nom	Modifié le	Type	Taille
code	28/05/2022 07:45	Dossier de fichiers	
envPySpark	27/05/2022 10:13	Dossier de fichiers	
envPySpark2	28/05/2022 10:05	Dossier de fichiers	
luncher	26/05/2022 18:05	Dossier de fichiers	
setup	26/05/2022 17:51	Dossier de fichiers	

Créer l'environnement envPySpark2

Conda create -y -p

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [version 10.0.19043.1706]
(c) Microsoft Corporation. Tous droits réservés.

(base) C:\Users\mariendiaye> conda create -y -p D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2_
```



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```
C:\WINDOWS\system32\cmd.exe
(base) C:\Users\mariendiaye> conda create -y -p D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2
Collecting package metadata (current_repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
  current version: 4.12.0
  latest version: 4.13.0

Please update conda by running

  $ conda update -n base -c defaults conda

## Package Plan ##

  environment location: D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2

Preparing transaction: done
Verifying transaction: done
Executing transaction: done
#
# To activate this environment, use
#
#   $ conda activate D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2
#
# To deactivate an active environment, use
#
#   $ conda deactivate

(base) C:\Users\mariendiaye>
```

Activer l'environnement qui vient d'être créé

```
C:\WINDOWS\system32\cmd.exe
(base) C:\Users\mariendiaye> activate D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>
```

Installer les packages openjdk, spark, findspark et pandas dans le nouvel environnement

Spark

Installer le package openjdk dans l'environnement : **conda install -y conda-forge::openjdk**

Installer le package pyspark dans l'environnement : **conda install -y -c conda-forge pyspark**



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```
C:\WINDOWS\system32\cmd.exe - conda install -y -c conda-forge pyspark
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>conda install -y -c conda-forge pyspark
Collecting package metadata (current_repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
  current version: 4.12.0
  latest version: 4.13.0

Please update conda by running

  $ conda update -n base -c defaults conda

## Package Plan ##

  environment location: D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2

  added / updated specs:
    - pyspark

The following NEW packages will be INSTALLED:

  ca-certificates    conda-forge/win-64::ca-certificates-2022.5.18.1-h5b45459_0
  intel-openmp       conda-forge/win-64::intel-openmp-2022.1.0-h57928b3_3787
  libblas            conda-forge/win-64::libblas-3.9.0-14_win64_mkl
  libcbblas          conda-forge/win-64::libcbblas-3.9.0-14_win64_mkl
  liblapack          conda-forge/win-64::liblapack-3.9.0-14_win64_mkl
  mkl                conda-forge/win-64::mkl-2022.0.0-h0e2418a_796
  numpy              conda-forge/win-64::numpy-1.21.6-py37h2830a78_0
  openssl            conda-forge/win-64::openssl-3.0.3-h8ffe710_0
  pandas             conda-forge/win-64::pandas-1.3.4-py37h9386db6_1
  pip                conda-forge/noarch::pip-22.1.1-pyhd8ed1ab_0

C:\WINDOWS\system32\cmd.exe - conda install -y -c conda-forge pyspark
The following NEW packages will be INSTALLED:

  ca-certificates    conda-forge/win-64::ca-certificates-2022.5.18.1-h5b45459_0
  intel-openmp       conda-forge/win-64::intel-openmp-2022.1.0-h57928b3_3787
  libblas            conda-forge/win-64::libblas-3.9.0-14_win64_mkl
  libcbblas          conda-forge/win-64::libcbblas-3.9.0-14_win64_mkl
  liblapack          conda-forge/win-64::liblapack-3.9.0-14_win64_mkl
  mkl                conda-forge/win-64::mkl-2022.0.0-h0e2418a_796
  numpy              conda-forge/win-64::numpy-1.21.6-py37h2830a78_0
  openssl            conda-forge/win-64::openssl-3.0.3-h8ffe710_0
  pandas             conda-forge/win-64::pandas-1.3.4-py37h9386db6_1
  pip                conda-forge/noarch::pip-22.1.1-pyhd8ed1ab_0
  py4j               pkgs/main/win-64::py4j-0.10.8.1-py37_0
  pyspark            conda-forge/win-64::pyspark-2.4.0-py37_1000
  python             conda-forge/win-64::python-3.7.12-h900ac77_100_cpython
  python-dateutil    conda-forge/noarch::python-dateutil-2.8.2-pyhd8ed1ab_0
  python_abi         conda-forge/win-64::python_abi-3.7-2_cp37m
  pytz               conda-forge/noarch::pytz-2022.1-pyhd8ed1ab_0
  setuptools         conda-forge/win-64::setuptools-59.8.0-py37h03978a9_1
  six                conda-forge/noarch::six-1.16.0-pyh6c4a22f_0
  sqlite             conda-forge/win-64::sqlite-3.38.5-h8ffe710_0
  tbb                conda-forge/win-64::tbb-2021.5.0-h2d74725_1
  ucrt               conda-forge/win-64::ucrt-10.0.20348.0-h57928b3_0
  vc                 conda-forge/win-64::vc-14.2-hb210afc_6
  vs2015_runtime     conda-forge/win-64::vs2015_runtime-14.29.30037-h902a5da_6
  wheel              conda-forge/noarch::wheel-0.37.1-pyhd8ed1ab_0

Preparing transaction: done
Verifying transaction: done
Executing transaction: done

(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>
```



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Findspark

conda install -y -c conda-forge findspark

```
C:\WINDOWS\system32\cmd.exe - conda install -y -c conda-forge pyspark - conda install -y -c conda-forge findspark
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>conda install -y -c conda-forge findspark
Collecting package metadata (current_repodata.json): done
Solving environment: done

==> WARNING: A newer version of conda exists. <==
  current version: 4.12.0
  latest version: 4.13.0

Please update conda by running

  $ conda update -n base -c defaults conda

## Package Plan ##

  environment location: D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2
  added / updated specs:
    - findspark

The following NEW packages will be INSTALLED:

  findspark          conda-forge/noarch::findspark-2.0.1-pyhd8ed1ab_0

Preparing transaction: done
Verifying transaction: done
Executing transaction: done

(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>_
```

Pandas

conda install -y -c anaconda pandas

Ajouter le nouvel environnement à jupyter notebook

pip install --user ipykernel



Enseignante : Pr. Marie NDIAYE

```
C:\WINDOWS\system32\cmd.exe - conda install -y -c conda-forge pyspark - conda install -y -c conda-forge findspark
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>pip install --user ipykernel
Requirement already satisfied: ipykernel in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (6.13.0)
Requirement already satisfied: matplotlib-inline>=0.1 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipykernel) (0.1.3)
Requirement already satisfied: ipython>=7.23.1 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipykernel) (7.33.0)
Requirement already satisfied: tornado>=6.1 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipykernel) (6.1)
Requirement already satisfied: traitlets>=5.1.0 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipykernel) (5.2.1.post0)
Requirement already satisfied: nest-asyncio in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipykernel) (1.5.5)
Requirement already satisfied: debugpy>=1.0 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipykernel) (1.6.0)
Requirement already satisfied: packaging in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipykernel) (21.3)
Requirement already satisfied: psutil in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipykernel) (5.9.1)
Requirement already satisfied: jupyter-client>=6.1.12 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipykernel) (7.3.1)
Requirement already satisfied: pickleshare in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (0.7.5)
Requirement already satisfied: decorator in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (5.1.1)
Requirement already satisfied: backcall in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (0.2.0)
Requirement already satisfied: setuptools>=18.5 in d:\working_directory\big_data\pyspark\envpyspark2\lib\site-packages (from ipython>=7.23.1->ipykernel) (59.8.0)
Requirement already satisfied: prompt-toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (3.0.29)
Requirement already satisfied: colorama in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (0.4.4)
Requirement already satisfied: pygments in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (2.12.0)
Requirement already satisfied: jedi>=0.16 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (0.18.1)
Requirement already satisfied: python-dateutil>=2.8.2 in d:\working_directory\big_data\pyspark\envpyspark2\lib\site-packages (from jupyter-client>=6.1.12->ipykernel) (2.8.2)
Requirement already satisfied: jupyter-core>=4.9.2 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from jupyter-client>=6.1.12->ipykernel) (4.10.0)
Requirement already satisfied: entrypoints in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from jupyter-client>=6.1.12->ip
```

```
C:\WINDOWS\system32\cmd.exe - conda install -y -c conda-forge pyspark - conda install -y -c conda-forge findspark
kernel) (59.8.0)
Requirement already satisfied: prompt-toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (3.0.29)
Requirement already satisfied: colorama in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (0.4.4)
Requirement already satisfied: pygments in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (2.12.0)
Requirement already satisfied: jedi>=0.16 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from ipython>=7.23.1->ipykernel) (0.18.1)
Requirement already satisfied: python-dateutil>=2.8.2 in d:\working_directory\big_data\pyspark\envpyspark2\lib\site-packages (from jupyter-client>=6.1.12->ipykernel) (2.8.2)
Requirement already satisfied: jupyter-core>=4.9.2 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from jupyter-client>=6.1.12->ipykernel) (4.10.0)
Requirement already satisfied: entrypoints in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from jupyter-client>=6.1.12->ipykernel) (0.4)
Requirement already satisfied: pyzmq>=22.3 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from jupyter-client>=6.1.12->ipykernel) (23.0.0)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from packaging->ipykernel) (3.0.9)
Requirement already satisfied: parso<0.9.0,>=0.8.0 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from jedi>=0.16->ipython>=7.23.1->ipykernel) (0.8.3)
Requirement already satisfied: pywin32>=1.0 in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from jupyter-core>=4.9.2->jupyter-client>=6.1.12->ipykernel) (304)
Requirement already satisfied: wcwidth in c:\users\mariendiaye\appdata\roaming\python\python37\site-packages (from prompt-toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0->ipython>=7.23.1->ipykernel) (0.2.5)
Requirement already satisfied: six>=1.5 in d:\working_directory\big_data\pyspark\envpyspark2\lib\site-packages (from python-dateutil>=2.8.2->jupyter-client>=6.1.12->ipykernel) (1.16.0)
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>
```

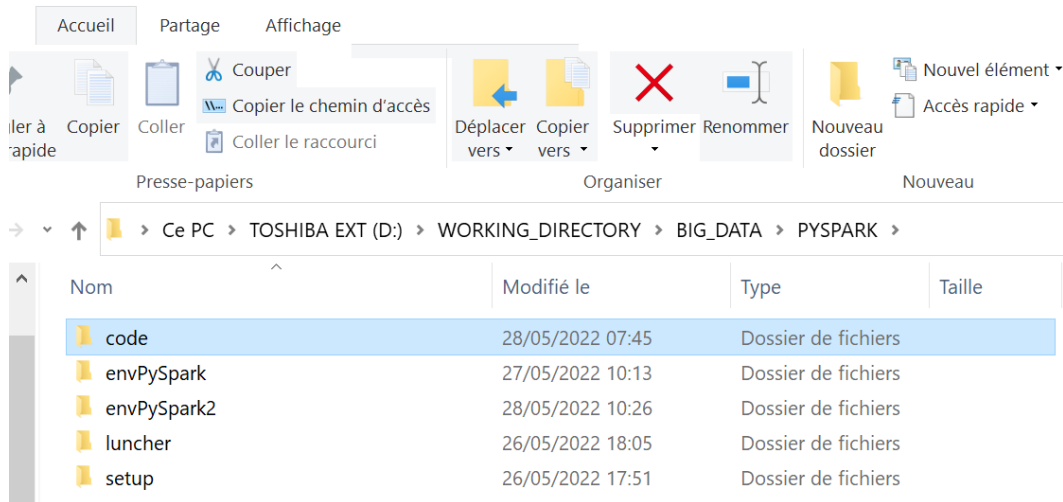
python -m ipykernel install --user --name=envPySpark2

```
C:\WINDOWS\system32\cmd.exe - conda install -y -c conda-forge pyspark - conda install -y -c conda-forge findspark
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye> python -m ipykernel install --user --name=envPySpark2
Installed kernelspec envPySpark2 in C:\Users\mariendiaye\AppData\Roaming\jupyter\kernels\envpyspark2
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>
```



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Créer un répertoire où vous allez mettre votre code



cd D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\code

```
C:\WINDOWS\system32\cmd.exe - conda install -y -c conda-forge pyspark - conda install -y -c conda-forge findspark
```

```
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>cd D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\code
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) C:\Users\mariendiaye>d:
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\code>
```

Lancer jupyter notebook

```
C:\WINDOWS\system32\cmd.exe - conda install -y -c conda-forge pyspark - conda install -y -c conda-forge findspark - jupyter notebook
```

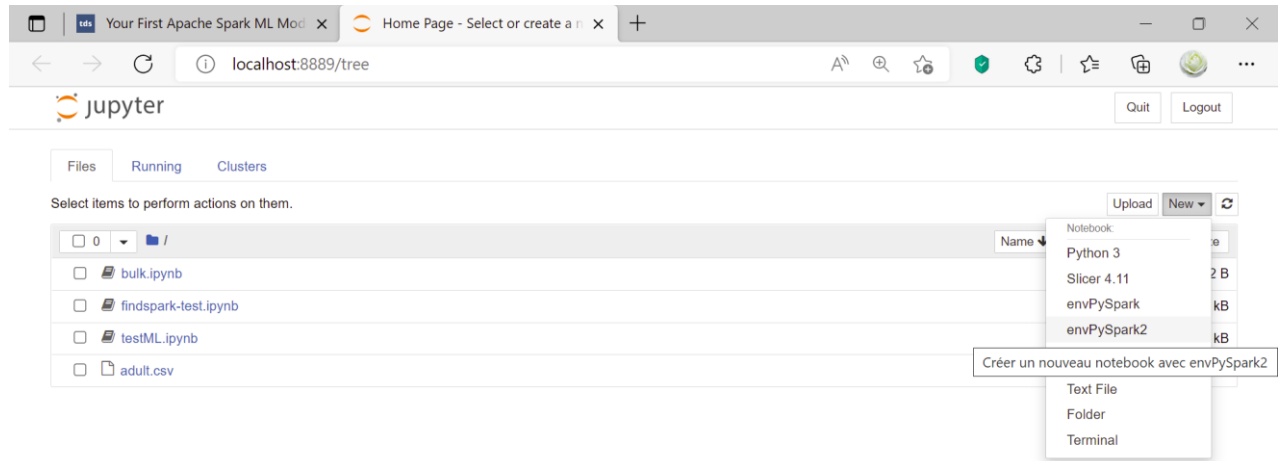
```
(D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\envPySpark2) D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\code>jupyter notebook
[I 10:38:35.217 NotebookApp] The port 8888 is already in use, trying another port.
[I 2022-05-28 10:38:38.307 LabApp] JupyterLab extension loaded from C:\ProgramData\Anaconda3\lib\site-packages\jupyterlab
[I 2022-05-28 10:38:38.307 LabApp] JupyterLab application directory is C:\ProgramData\Anaconda3\share\jupyter\lab
[I 10:38:38.317 NotebookApp] Serving notebooks from local directory: D:\WORKING_DIRECTORY\BIG_DATA\PYSPARK\code
[I 10:38:38.317 NotebookApp] Jupyter Notebook 6.3.0 is running at:
[I 10:38:38.317 NotebookApp] http://localhost:8889/?token=b28502658e2aec8de9e9a14586f5489589da8922e06afe7c
[I 10:38:38.317 NotebookApp] or http://127.0.0.1:8889/?token=b28502658e2aec8de9e9a14586f5489589da8922e06afe7c
[I 10:38:38.317 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 10:38:38.697 NotebookApp]
```

```
To access the notebook, open this file in a browser:
file:///C:/Users/mariendiaye/AppData/Roaming/jupyter/runtime/nbserver-11900-open.html
Or copy and paste one of these URLs:
http://localhost:8889/?token=b28502658e2aec8de9e9a14586f5489589da8922e06afe7c
or http://127.0.0.1:8889/?token=b28502658e2aec8de9e9a14586f5489589da8922e06afe7c
```



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Créer un nouveau fichier



Mettre le code ci-dessous et exécuter.

```
import pyspark
pyspark. version
```

Mettre les codes ci-dessous et exécuter à nouveau.

```
import findspark
findspark.init()
```

```
from pyspark.sql import SparkSession as sps
spark = sps.builder.master("local[1]").appName("MySpakExample.com").getOrCreate()
```

```
rdd = spark.sparkContext.parallelize([1,2,3,4,5])
rdd.count()
```

Utiliser PySpark

Faire les deux exemples ci-dessous du tutoriel accessible via l'adresse :

<https://sparkbyexamples.com/pyspark-tutorial/>

```
# Create RDD from parallelize
dataList = [("Java", 20000), ("Python", 100000), ("Scala", 3000)]
rdd=spark.sparkContext.parallelize(dataList)

# Create RDD from external Data source
rdd2 = spark.sparkContext.textFile("/path/test.txt")
```



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```
#Using createDataFrame()
data = [('James', '', 'Smith', '1991-04-01', 'M', 3000),
        ('Michael', 'Rose', '', '2000-05-19', 'M', 4000),
        ('Robert', '', 'Williams', '1978-09-05', 'M', 4000),
        ('Maria', 'Anne', 'Jones', '1967-12-01', 'F', 4000),
        ('Jen', 'Mary', 'Brown', '1980-02-17', 'F', -1)
]
columns = ["firstname", "middlename", "lastname", "dob", "gender", "salary"]
df = spark.createDataFrame(data=data, schema = columns)
df.printSchema()
df.show()
```

Le jeu de données utilisé ([student_data_bis.csv](#)) est téléchargeable à partir de Kaggle (<https://www.kaggle.com/datasets/goyaladi/iit-admissions-dataset>). Vous pouvez utiliser un autre fichier csv de votre choix si le téléchargement est difficile.

```
#DataFrame from external data sources
df1 = spark.read.option("delimiter", ";").option("header",
"true").csv("E:/2023/datasets/Students/student_data_bis.csv")
df1.printSchema()
df1.show()

##https://sparkbyexamples.com/pyspark/pyspark-read-csv-file-into-
dataframe/
#df = spark.read.csv("E:/2023/datasets/Students/student_data.csv")
#df.printSchema()
```

Lire la documentation de PySark

<https://spark.apache.org/docs/latest/api/python/>

<https://sparkbyexamples.com/pyspark/install-pyspark-in-anaconda-jupyter-notebook/>